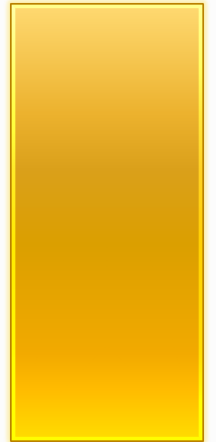


TEXT MINING

L01. INTRODUCTION



SUZAN VERBERNE 2024

TODAY'S LECTURE

- Introduction
- What is text mining
- Example Text Mining projects
- Challenges of text data
- The Text Mining pipeline
- Text Mining models
- Case (exercise)
- Structure of this course

COURSE INFORMATION

- Brightspace page:
<https://brightspace.universiteitleiden.nl/d2l/home/331586>
- Course web page: <http://tmr.liacs.nl/TM.html>
- Lectures: Wednesday morning, 11.00-12.45, GORL / 03

CONTACT INFORMATION

- Prof. dr. Suzan Verberne
<http://liacs.leidenuniv.nl/~verbernes/>
- Teaching assistants:
 - Amin Abolghasemi (PhD student)
 - Yumeng Wang (PhD student)
 - Hedieh Pourghasem (master student)
 - Norika Narimatsu (master student)
 - Tiago Cumetti (master student)
 - Dimitrios Kourntidis (master student)
- Contact: tmcourse@liacs.leidenuniv.nl

YOUR PROGRAM

Quick round (raise hands): what is your master program/specialisation?

- Computer Science: Data Science
- Computer Science: Artificial Intelligence
- Computer Science: Bioinformatics
- Computer Science: other
- Media Technology
- ICT in Business and the Public Sector
- Other

YOUR BACKGROUND KNOWLEDGE

Quick round (raise hands)

- Who has taken a course in Machine Learning?
- Who has taken the course Information Retrieval?
- Who has taken the course Introduction to Deep Learning last year?
- Who is currently taking the course Introduction to Deep Learning?
- Who can program in Python?
- Who knows what a vector is?
- Who knows what a named entity is?
- Who has heard of BERT?
- Who has heard of GPT?



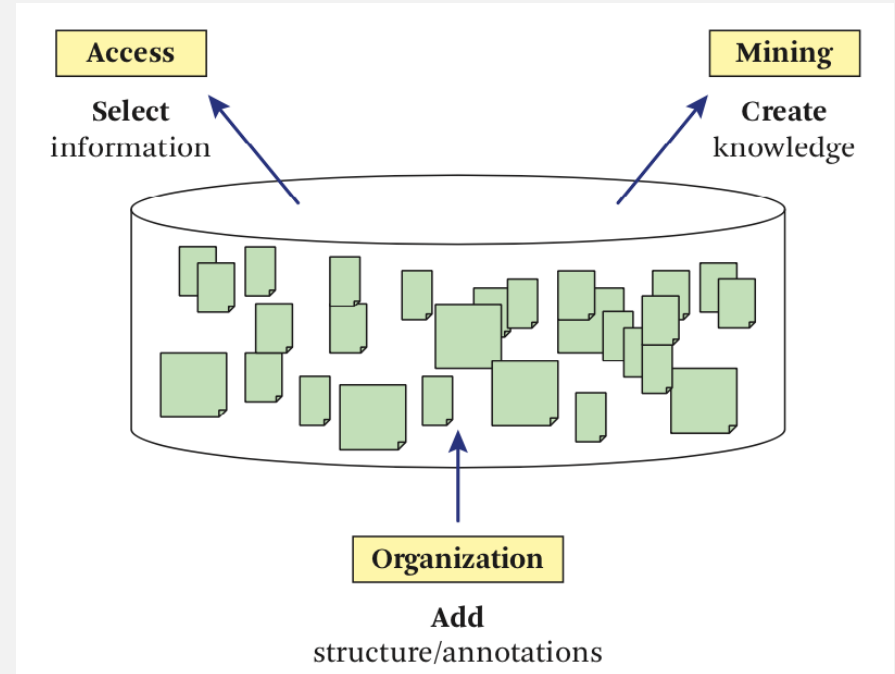
WHAT IS TEXT MINING

WHY TEXT MINING?

- A large portion of the world's knowledge is stored in text:
 - web pages
 - user-generated content on the web (social media)
 - electronic health records
 - scientific literature
 - patents
 - political/legal texts

WHAT IS TEXT MINING

- Text mining: Automatic extraction of knowledge from text
- Text = unstructured
- Knowledge = structured



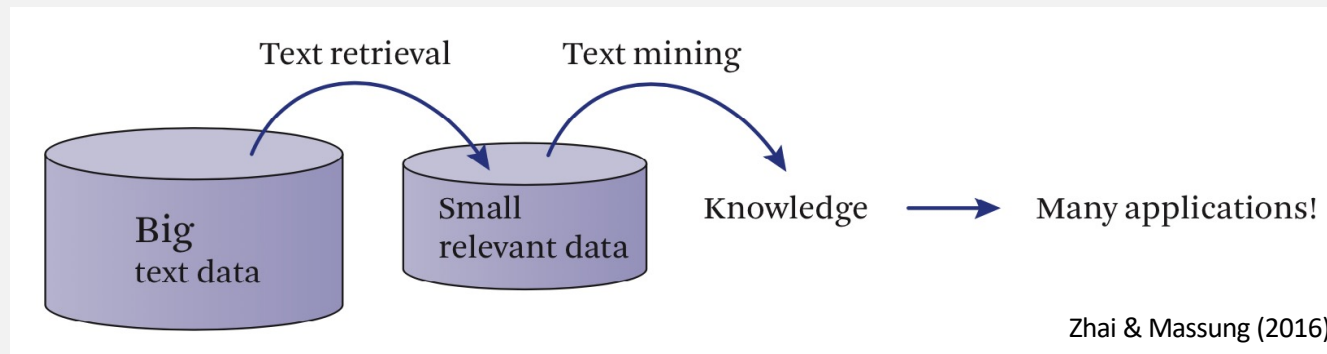
Zhai & Massung (2016)

TEXT MINING AND NLP

- **NLP = Natural Language Processing**
- Text Mining applications use NLP methods
- NLP is a large and active research field
 - NLP has a fundamental component (**computational linguistics**)
 - But in its current state, it is a core-AI/machine learning field
- Not all NLP tasks are text mining tasks
 - E.g. machine translation, speech recognition, semantic parsing

TEXT MINING AND INFORMATION RETRIEVAL

- Text Mining (TM) and Information Retrieval (IR) are related disciplines
- In many applications, **IR is the first step of the TM process**
 - First retrieve documents (IR), then extract and structure the relevant information



- Core aspects of IR that are not common in NLP/TM:
 - ranking models
 - user modelling and evaluation

EXAMPLE TEXT MINING PROJECTS

EXAMPLE TEXT MINING PROJECTS

- **Classifying** political election manifestos
- Finding relevant **names** (persons, organisations) in political speeches
- The automatic labelling of medical **entities** in Dutch endoscopy reports
- Finding biomedical **entities** (genes, proteins, diseases) in Chinese patents
- Finding archaeological **entities** (locations, artefacts, time periods) in archaeological reports
- **Summarization** of long discussion threads
- Analyzing the political **positioning** of newspapers
- **Sentiment** analysis on patient experience survey data

FINDING NAMES IN POLITICAL SPEECHES

- **Goal:** find locations, persons, and organizations mentioned in recordings of European Parliament Committee meetings
- **Data:** output of automatic speech recognition (ASR):
 - SO A LEX DOID BY ACCLAMATION CONGRATULATION STERMORISE I
DECLARE UNTE CANDIDATE ELACTED BY ACCLAMATION AND ASKED ATHE
NEW CHAIR TO TAKE SID
- **Challenges:**
 - ASR errors
 - No punctuation and capitalisation
 - No labelled data

INFORMATION EXTRACTION FROM CHINESE BIOMEDICAL PATENTS

➤ **Goal:** extract genes, proteins, and diseases from Chinese biomedical patents

➤ **Data** example:

➤ **Challenges:**

➤ Complex patent language

➤ Multi-lingual texts ('code-switching')

➤ OCR* errors

1. 一种分离的抗体或其抗原结合片段,其特异性结合赖氨酰氧化酶样 (LOXL2) 蛋白,并包含重链可变区互补决定区 (CDR) 和轻链可变区 CDR,所述重链可变区 CDR 包含如 SEQ ID NO: 41、42 和 43 所示的氨基酸序列,以及所述轻链可变区 CDR 包含如 SEQ ID NO: 57、58 和 59 所示的氨基酸序列。

2. 权利要求 1 的抗体或其抗原结合片段,其包含与 SEQ ID NO: 1 所示氨基酸序列具有至少 75% 氨基酸序列相同性的可变重链。

3. 权利要求 1 或 2 的抗体或其抗原结合片段,其包含与 SEQ ID NO: 2 所示氨基酸序列具有至少 75% 氨基酸序列相同性的可变轻链。

4. 权利要求 1-3 任一项的抗体或其抗原结合片段,其包含与 SEQ ID NO: 1 所示氨基酸序列具有至少 75% 氨基酸序列相同性的可变重链和与 SEQ ID NO: 2 所示氨基酸序列具有至少 75% 氨基酸序列相同性的可变轻链。

5. 权利要求 1-4 任一项的抗体,所述抗体特异性结合 LOXL2 的结合亲和力是结合 LOX、LOXL1、LOXL3 或 LOXL4 中至少一种的至少 2、5、10、50、100、500 或 1000 倍。

INFORMATION EXTRACTION FROM DUTCH ARCHAEOLOGICAL REPORTS

- **Goal:** to automatically label a large archaeological text collection (~658 million words) with relevant entities

Entity	Description	Examples
Artefact	An archaeological object found in the ground.	Axe, pot, stake, arrow head, coin
Time Period	A defined (archaeological) period in time.	Middle Ages, Neolithic, 500 BC, 4000 BP
Location	A placename or (part of) an address.	Amsterdam, Steenstraat 1, Lutjebroek
Context	An anthropogenic, definable part of a stratigraphy. Something that can contain Artefacts	Rubbish pit, burial mound, stake hole
Material	The material an Artefact is made of.	Bronze, wood, flint, glass
Species	A species' name (in Latin or Dutch)	Cow, <i>Corvus Corax</i> , oak

- **Challenges:**

- **Synonymy:** there are numerous ways to write concepts relevant to archaeology. E.g. 'Early Middle Ages' = 'Early Medieval Period' = 'Merovingian Period' = '600 CE' = '1400 BP'
- **Polysemy:** one word can have multiple meanings. E.g. Swifterbant = a location, a type of pottery, and a time period.

SENTIMENT ANALYSIS ON PATIENT EXPERIENCE SURVEY DATA

- **Goal:** identify positive and negative aspects in patient survey after they have left the hospital.
- **Data** examples:

What went well? (=positive)	What could be better? (=negative)
Good control by the doctors	The long waiting time
the surgery according to the doctor	Better information about the post-pain
I was very well taken care of by the nursing staff	the information about self-medication at home after discharge is very unclear

- **Challenges:**
 - Noise in the text: abbreviations, typos, spelling errors
 - Noise in the labels: positive comments in the negative field
 - Very short texts are difficult to classify

CHALLENGES OF TEXT DATA

1. TEXT DATA IS UNSTRUCTURED

➤ Or at best semi-structured:

- PHYSICAL EXAMINATION: On physical examination, her blood pressure was 104/73, pulse 79. In general, she was a woman in no acute distress. HEENT: Nonicteric. Pupils are equal, round, and reactive to light. Extraocular movements are full. Pharynx is benign. Tongue midline. Neck is supple.



Only Murders in the Building is an American **mystery comedy-drama** television series created by **Steve Martin** and **John Hoffman**. The main plot focuses on a trio of strangers (played by Steve Martin, **Martin Short**, and **Selena Gomez**), all with a shared interest in **true crime** podcasts, who become friends while investigating a succession of suspicious murders in the Arconia, their affluent **Upper West Side** apartment building, and producing their own podcast about the cases, titled *Only Murders in the Building*. Its three ten-episode seasons premiered on **Hulu** in August 2021, June 2022, and August 2023. In October 2023, it was renewed for a fourth season which premiered on August 27, 2024.

2. TEXT DATA IS NOISY

Types of noise:

- Noise in the **source**:

- Non-content information in the data (e.g. HTML/CSS/javascript, URLs)
- Encoding errors, line endings

- Noise in the **content**:

- OCR errors (in scanned text), spelling errors and typos
- Formatting (lost when converted to plain text)

- Noise in the **labels**:

- Errors or incomplete



```

48 <div class="search-container">
49 <form action="https://studiegids.universiteitleidennl/en/search" class="search-form wrapper">
50 <input type="hidden" name="for" value="courses" />
51 <div class="search">
52 <!-- <label class="select">
53 <span>Search for</span>
54 <select name="for">
55 <option value="courses">Courses</option>
56 <option value="programmes">Programmes</option>
57 </select>
58 </label> -->
59 <label class="keywords no-label">
60 <span>Keywords</span>
61 <input type="search" name="q" value="" placeholder="What are you searching for?" autofocus autocomplete="off"/>
62 </label> <label class="select no-label">
63 <span>Academic year</span>
64 <select name="edition">
65 <option>2025-2026</option>
66 <option selected>2024-2025</option>
67 <option>2023-2024</option>
68 <option>2022-2023</option>
69 <option>2021-2022</option>
70 <option>2020-2021</option>
71 <option>2019-2020</option>
72 <option>2018-2019</option>
73 <option>2017-2018</option>
74 <option>2016-2017</option>
75 <option>2015-2016</option>
76 <option>2014-2015</option>
77 <option>2013-2014</option>
78 <option>2012-2013</option>
79 <option>2011-2012</option>
80 <option>2010-2011</option>
81 </select>
82 </label> <button type="submit">Search</button> </div>
83 </form>
84 </div>
85
86
87
88 <div class="main-container">
89 <div class="wrapper summary">
90 <article class="content">
91 <h1>Text Mining</h1>
92 <div class="meta">
93 <div class="entity">
94 <span><b>Course</b></span>
95 </div>
96 <div class="edition">
97 <span>2024-2025</span>

```



RESUMO – A preservação e a geração de porosidade em reservatórios clásticos profundos são controladas por diversos processos e situações geológicas específicas. Os principais fatores de preservação de porosidade são os seguintes: 1)- soterramento tardio do reservatório à sua atual profundidade; 2)- desenvolvimento de pressões anormais de fluidos; 3)- estabilidade composicional dos grãos do arcabouço; 4)- recobrimento dos grãos por cutículas ou franjas de argilas e/ou óxidos; 5)- cimentação precoce parcial por carbonatos ou sulfatos; e 6)- saturação precoce do reservatório por hidrocarbonetos. Os processos e solventes para a geração de porosidade em subsuperfície são estes: 1)- infiltração profunda de águas meteóricas; 2)- CO₂ da maturação térmica da matéria orgânica; 3)- solventes orgânicos (principalmente ácidos carboxílicos) liberados pela matéria orgânica; 4)- fluidos ácidos de reações inorgânicas com argilominerais; 5)- redução termogênica de sulfato por hidrocarbonetos.

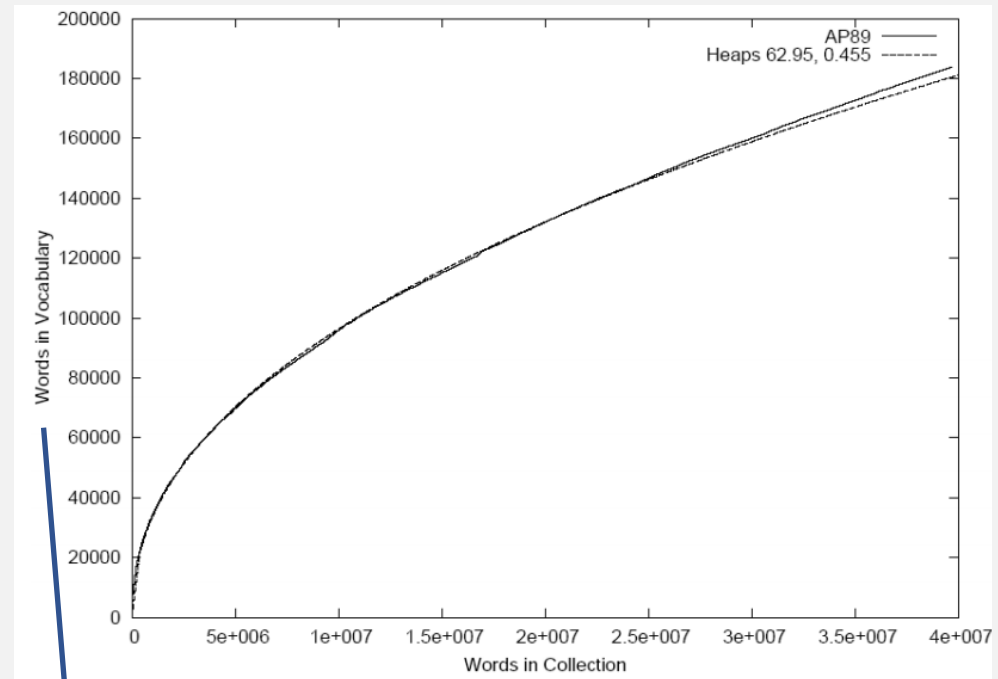
(a) Original PDF Document

RESUMO - A preservação e a geração de porosidade em reservat6rios elásticos profundos são controladas por diversos processos e situações geol6gicas especflicas. Os principais fatores de preservação de porosidade são os seguintes: 1)- soterramento tardio do reservat6rio à sua atual profundidade; 2)- desenvolvimento de pressões anormais de fluidos; 3)- estabilidade composicional dos grãos do arcabouço; 4)- recobrimento dos grãos por cutrculas ou franjas de argilas e/ou 6xidos; 5)- cimentação precoce parcial por carbonatos ou sulfatos; e 6)- saturação precoce do reservat6rio por hidrocarbonetos. Os processos e solventes para a geração de porosidade em subsuperffcie são estes: 1)- infiltração profunda de águas mete6ricas; 2)- CO₂ da maturação térmica da matéria orgânica; 3)- solventes orgânicos (principalmente ácidos carboxnicos) liberados pela matéria orgânica; 4)- fluidos ácidos de reações inorgânicas com argilominerais; 5)- redução termogênica de sulfato por hidrocarbonetos,

(b) Extracted Textual Contents

3. LANGUAGE IS INFINITE

- A new document in your collection is likely to add new terms
- The number of new words will increase very rapidly when the corpus is small and would continue to increase indefinitely, but at a slower rate for larger corpus (**Heaps' Law**)



Vocabulary size

4. LANGUAGE IS AMBIGUOUS

2/5 6 months ago on  Google



Hotel more in the 2** standard

Room and breakfast correct.

The only plus is the location.

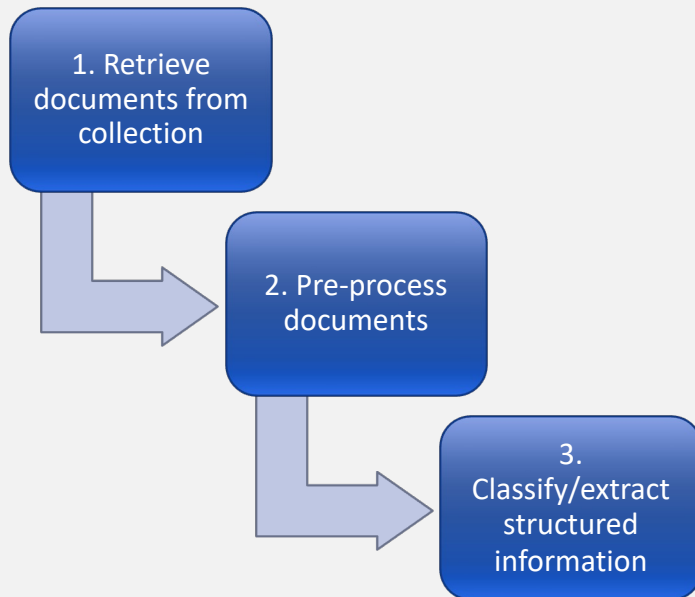
Service not very helpful. When asked where to park a car with a roof box (over 2 m high) he replied "I don't know"

After leaving the hotel, I found a safe parking place by myself Approx 50 m from the hotel.

- 'the 2** standard' (poor on Google Maps, but would be excellent in Michelin)
- 'Room and breakfast correct' (but suggests it is minimal)
- 'Safe parking space' (sounds great) '50m from hotel' (very close)

THE TEXT MINING PIPELINE

THE TEXT MINING PIPELINE



Example TM problem: estimate the level of support on social media for the new government

1. **IR**: retrieve social media messages that are about the new government
2. **Pre-processing**: Filter duplicates. Clean from noise. Anonymize if necessary
3. **NLP**: classify all messages in pro/against/neutral with respect to the new government

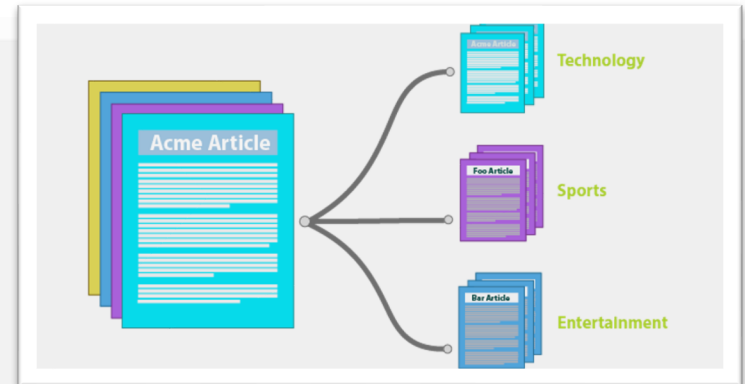
TYPES OF TEXT PROCESSING TASKS

We distinguish three types of text mining tasks:

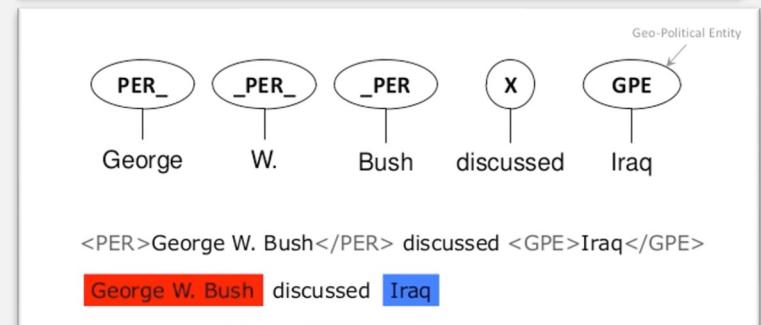
1. **Text classification/clustering**: assign a category or cluster per document
 - the 'document' can be any text type (newspaper article, tweet, e-mail, text message, patent, even a single sentence)
 - the 'category' can be any type of label (topic, relevance/importance, author, sentiment, ...)
2. **Sequence labelling**: assign a category per word in a text
 - e.g. identify person names and places in a text (named entity recognition)
3. **Text-to-text generation**: input is text, output is text
 - summarization, translation

TYPES OF TEXT PROCESSING TASKS

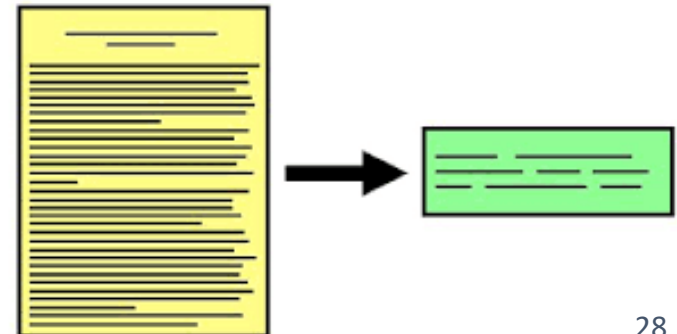
1. Text classification



2. Named entity recognition (= sequence labelling)



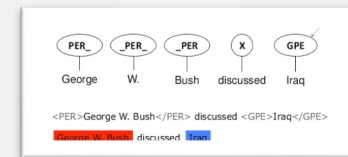
3. Summarization (= sequence-to-sequence)



TEXT MINING MODELS

TEXT AS CLASSIFICATION OBJECT

- Important distinction: Text as **classification object** vs. text as sequence

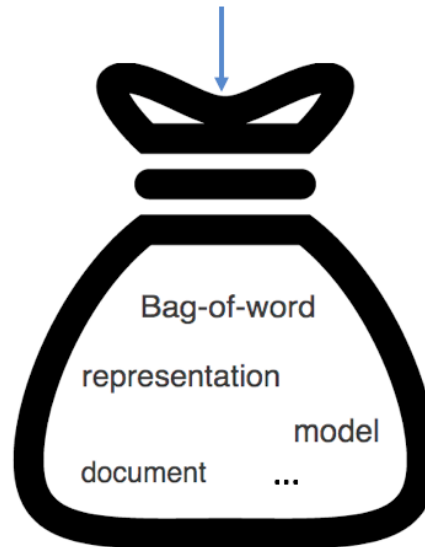


- When we classify text as a whole, the **word order is less relevant** than when we do sequence labelling
- Traditional **text classification methods** represent the text as a '**bag of words**'
- In the bag-of-words model, each word in the collection becomes a feature

TEXT AS CLASSIFICATION OBJECT

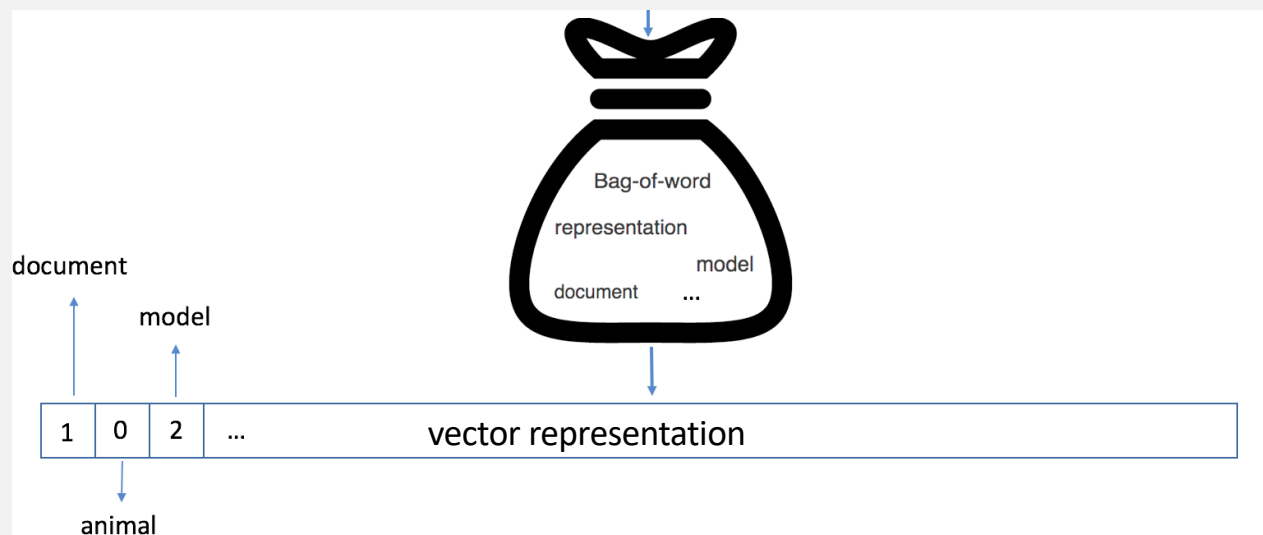
➤ The bag of words:

Bag-of-word model is an orderless document representation. If a document contains the words "I like movies too", the bag-of-words representation will not retain the order of the words. A vector space model can be used to store this spatial information with the same term frequency of each unit as before.



TEXT AS CLASSIFICATION OBJECT

- Traditional Bag-of-words model:
 - Word order is not relevant
 - Punctuation is not relevant
 - Sentence and paragraph borders are not relevant

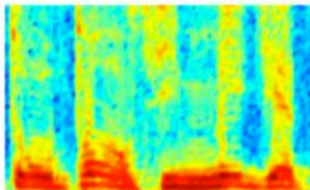


TEXT AS CLASSIFICATION OBJECT

- When we use words as features:
 - Each term in the collection becomes a dimension in the vector space
 - Only a few of all words occur in a given document
 - Hence, word vectors are high-dimensional, sparse vectors

>10,000 dimensions

AUDIO



Audio Spectrogram

DENSE

IMAGES

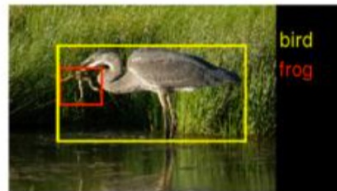
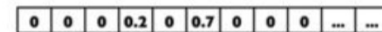


Image pixels

DENSE

TEXT

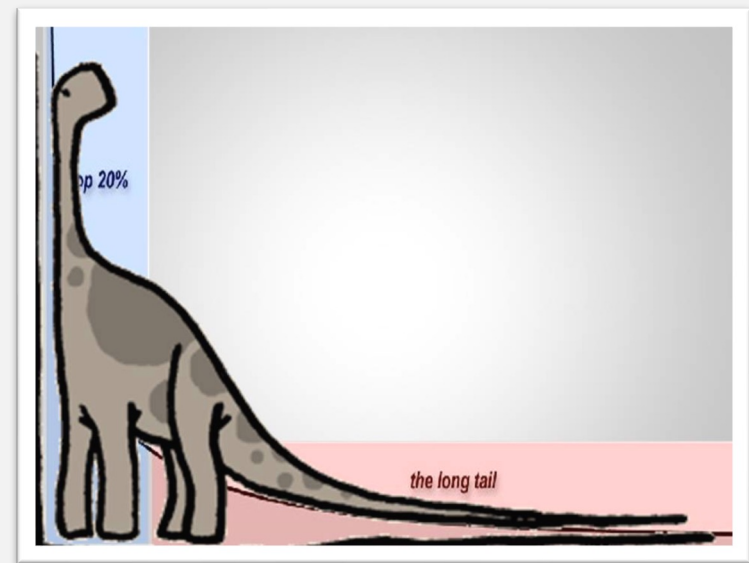
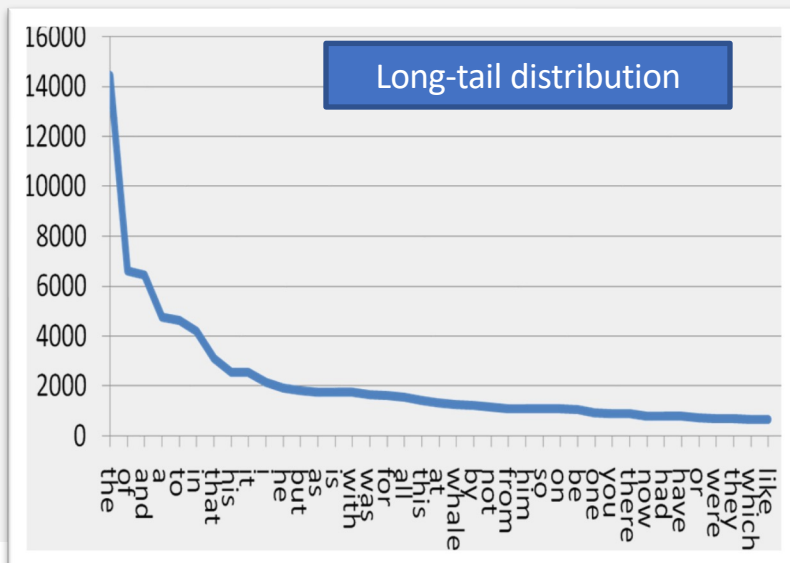


Word, context, or document vectors

SPARSE

ZIPF'S LAW

- Given a text collection, the frequency of a word is inversely proportional to its rank in the frequency table (**Zipf's Law**): word frequency $\propto \frac{1}{\text{word rank}}$
- In English, the top four most frequent words are about 10-15% of all word occurrences. The top 50 words are 35-40% of word occurrences.



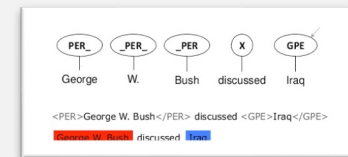
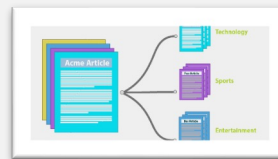
DENSE REPRESENTATIONS FOR TEXT

- Alternative to words as features: **word embeddings**
 - words are represented as vectors
 - the vector space is lower-dimensional
 - the vector space is dense
 - the dimensions are latent (are not individually interpretable) and learnt from data
 - similar words are close to each other in the space
- More details in lecture 3 and further

100-800 dimensions

TEXT AS SEQUENTIAL DATA

- Important distinction: Text as classification object vs. text as **sequence**



- If we want to **extract knowledge from text**, sequential information matters:
 - word order (sequence)
 - punctuation
 - capitalisation

TEXT AS SEQUENTIAL DATA

- E.g. names, dates, and titles from biographical text:

Daisy Jazz Isobel Ridley (born 10 April 1992) is an English actress who rose to international prominence through playing the role of Rey in the Star Wars sequel trilogy: The Force Awakens (2015), The Last Jedi (2017), and The Rise of Skywalker (2019).

- E.g. medications and side effects from patient experiences:

Since I started on **Gleevec**, I **can't fall asleep** at all.

TRANSFORMER MODELS

TRANSFORMER MODELS

- Most of the current NLP methods are based on the **Transformer** architecture, proposed in 2017 by Google research

[PDF] **Attention is all you need**

[A Vaswani](#) - Advances in Neural Information Processing Systems, 2017 - [user.phil.hhu.de](#)

Attention is all you need **Attention is all you need** ...

☆ Save  Cite Cited by 131317 Related articles 

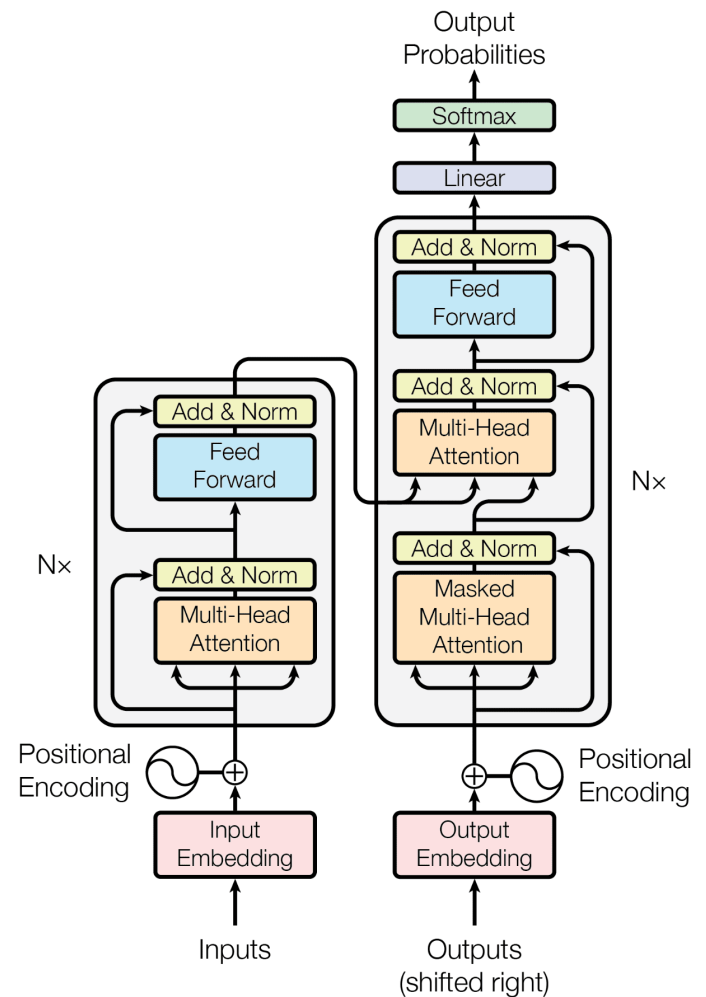
TRANSFORMER MODELS

- Transformers were designed for **sequence-to-sequence** tasks
- Sequence-to-sequence = text in, text out
- E.g. translation

The screenshot displays the Google Translate interface. On the left, the 'Detect language' dropdown is set to 'English'. The input text reads: 'Zipf's law (/zɪf/, German: [tsɪpf]) is an empirical law that often holds, approximately, when a list of measured values is sorted in decreasing order. It states that the value of the nth entry is inversely proportional to n.' Below the text are icons for voice input and output, and a character count of '226 / 5,000'. On the right, the 'English' dropdown is set to 'Dutch'. The translated text reads: 'De wet van Zipf (/zɪf/, Duits: [tsɪpf]) is een empirische wet die vaak, bij benadering, geldt wanneer een lijst met gemeten waarden in afnemende volgorde wordt gesorteerd. Deze wet stelt dat de waarde van de n-de invoer omgekeerd evenredig is met n.' Below the translation are icons for voice input/output, a star for favorites, and a share icon.

TRANSFORMER MODELS

- The transformer architecture has two components:
 - **Encoders:** process the input text
 - **Decoders:** generate the output text



TRANSFORMER MODELS: BERT

Bert: Pre-training of deep bidirectional transformers for language understanding

J Devlin, [MW Chang](#), [K Lee](#), [K Toutanova](#) - arXiv preprint arXiv ..., 2018 - [arxiv.org](#)

... **BERT** and its detailed implementation in this section. There are two steps in our framework: **pre-training** and fine... of **BERT** by evaluating two **pretraining** objectives using exactly the same ...

☆ Save 📄 Cite Cited by 110824 Related articles All 46 versions ⇔



- BERT (Bidirectional Encoder Representations from Transformers)
 - An **encoder**-only transformer
 - Input is text, output is embeddings
- A growing family of models since 2018:

<https://www.aclweb.org/anthology/N19-1423.pdf>

TRANSFORMER MODELS: BERT

The screenshot shows the Hugging Face website interface. At the top, the Hugging Face logo is on the left, and navigation links for Models, Datasets, Spaces, Posts, Docs, Solutions, Pricing, Log In, and Sign Up are on the right. A search bar contains the text 'bert'. Below the navigation bar, the left sidebar shows various task categories: Multimodal (Image-Text-to-Text, Visual Question Answering, Document Question Answering, Video-Text-to-Text), Computer Vision (Depth Estimation, Image Classification, Object Detection, Image Segmentation, Text-to-Image, Image-to-Text, Image-to-Image, Image-to-Video, Unconditional Image Generation, Video Classification, Text-to-Video, Zero-Shot Image Classification, Mask Generation, Zero-Shot Object Detection, Text-to-3D, Image-to-3D, Image Feature Extraction), and Natural Language Processing (Text Classification, Token Classification, Table Question Answering, Question Answering, Zero-Shot Classification, Translation, Summarization, Feature Extraction). The main content area displays a list of models under the 'Models' tab, with a search filter 'bert' applied. The list includes models like 'google-bert/bert-base-uncased', 'google-bert/bert-base-chinese', 'dslim/bert-base-NER', 'nlpTown/bert-base-multilingual-uncased-sentiment', 'pySentimiento/bertweet-pt-sentiment', 'google-bert/bert-base-cased', 'google-bert/bert-base-multilingual-uncased', 'google-bert/bert-large-cased', 'google-bert/bert-large-uncased', 'CAMEL-Lab/bert-base-arabic-camelbert-da-sentiment', and 'CAMEL-Lab/bert-base-arabic-camelbert-mix'. Each model entry shows its name, task type, update date, size, and popularity metrics.

Hugging Face Search models, datasets, users...

Models 30,209

Full-text search 1 Sort: Trending

Tasks Libraries Datasets Languages Licenses Other

Filter Tasks by name

Multimodal

- Image-Text-to-Text
- Visual Question Answering
- Document Question Answering
- Video-Text-to-Text

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Text-to-Image
- Image-to-Text
- Image-to-Image
- Image-to-Video
- Unconditional Image Generation
- Video Classification
- Text-to-Video
- Zero-Shot Image Classification
- Mask Generation
- Zero-Shot Object Detection
- Text-to-3D
- Image-to-3D
- Image Feature Extraction

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Feature Extraction

Models

- google-bert/bert-base-uncased**
Fill-Mask • Updated Feb 19 • 59.9M • 1.76k
- google-bert/bert-base-chinese**
Fill-Mask • Updated Feb 19 • 1.83M • 920
- dslim/bert-base-NER**
Token Classification • Updated Jan 25 • 5.55M • 472
- nlpTown/bert-base-multilingual-uncased-sentiment**
Text Classification • Updated Jul 27, 2023 • 1.38M • 297
- pySentimiento/bertweet-pt-sentiment**
Updated Mar 29, 2023 • 17.4k • 8
- google-bert/bert-base-cased**
Fill-Mask • Updated Feb 19 • 5.69M • 247
- google-bert/bert-base-multilingual-uncased**
Fill-Mask • Updated Feb 19 • 583k • 98
- google-bert/bert-large-cased**
Fill-Mask • Updated Feb 19 • 485k • 30
- google-bert/bert-large-uncased**
Fill-Mask • Updated Feb 19 • 884k • 108
- CAMEL-Lab/bert-base-arabic-camelbert-da-sentiment**
Text Classification • Updated Oct 17, 2021 • 38.7k • 39
- CAMEL-Lab/bert-base-arabic-camelbert-mix**
Fill-Mask • Updated Sep 14, 2021 • 1.85k • 12

<https://huggingface.co/models?search=bert>



TRANSFORMER MODELS: GPT

[PDF] Improving language understanding by generative pre-training

[A Radford - 2018 - hayate-lab.com](#)

Improving language understanding by generative pre-training Improving language understanding by generative pre-training ...

☆ Save 📄 Cite Cited by 10556 Related articles 🔗

- GPT (Generative Pre-trained Transformer)
 - A **decoder**-only transformer
 - Input is prompt; output is text
- A growing family of models since 2018:
 - GPT-2, GPT-3, ChatGPT, GPT-4o, BLOOM, Llama, Mistral, ...

TRANSFORMER MODELS: GPT

The screenshot displays the Hugging Face homepage. At the top, the 'Hugging Face' logo is on the left, and navigation links for 'Models', 'Datasets', 'Spaces', 'Posts', 'Docs', 'Solutions', 'Pricing', 'Log In', and 'Sign Up' are on the right. A search bar is positioned below the logo. The left sidebar contains a 'Tasks' section with filters for 'Multimodal' (Image-Text-to-Text, Visual Question Answering, Document Question Answering, Video-Text-to-Text) and 'Computer Vision' (Depth Estimation, Image Classification, Object Detection, Image Segmentation, Text-to-Image, Image-to-Text, Image-to-Image, Image-to-Video, Unconditional Image Generation, Video Classification, Text-to-Video, Zero-Shot Image Classification, Mask Generation, Zero-Shot Object Detection, Text-to-3D, Image-to-3D, Image Feature Extraction). Below this is the 'Natural Language Processing' section with filters for Text Classification, Token Classification, Table Question Answering, Question Answering, Zero-Shot Classification, Translation, Summarization, Feature Extraction, Text Generation, Text2Text Generation, Fill-Mask, and Sentence Similarity. The main content area is titled 'Models 135,020' and includes a 'Filter by name' input. It displays a grid of model cards for text generation. Each card shows the model name, its primary task, update frequency, and performance metrics (downloads, likes, and reviews).

Model Name	Task	Updated	Downloads	Likes	Reviews
meta-llama/Meta-Llama-3.1-8B-Instruct	Text Generation	Updated 15 days ago	2.88M	2.21k	
CausalLM/miniG	Text Generation	Updated 3 days ago	275k	178	
CohereForAI/c4ai-command-r-08-2024	Text Generation	Updated about 14 hours ago	792	109	
CohereForAI/c4ai-command-r-plus-08-2024	Text Generation	Updated 5 days ago	387	91	
microsoft/Phi-3.5-mini-instruct	Text Generation	Updated 5 days ago	137k	418	
meta-llama/Meta-Llama-3.1-8B	Text Generation	Updated Jul 24	492k	701	
ai21labs/AI21-Jamba-1.5-Mini	Text Generation	Updated 3 days ago	18.6k	189	
speakeash/Bielik-11B-v2.2-Instruct	Text Generation	Updated 7 days ago	784	38	
microsoft/Phi-3.5-MoE-instruct	Text Generation	Updated 5 days ago	26.2k	440	
ai21labs/AI21-Jamba-1.5-Large	Text Generation	Updated 3 days ago	998	156	
speakeash/Bielik-11B-v2	Text Generation	Updated 7 days ago	202	29	
meta-llama/Meta-Llama-3.1-70B-Instruct	Text Generation	Updated 14 days ago	782k	431	
bartowski/c4ai-command-r-08-2024-GGUF	Text Generation	Updated 5 days ago	3.32k	29	
google/gemma-2-9b-it	Text Generation	Updated 7 days ago	1.92M	425	
meta-llama/Meta-Llama-3-8B	Text Generation	Updated May 13	2.11M	5.58k	
mistralai/Mistral-Nemo-Instruct-2407	Text Generation	Updated 15 days ago	310k	1.01k	
meta-llama/Meta-Llama-3-8B-Instruct	Text Generation	Updated May 28	1.83M	6.33k	
meta-llama/Meta-Llama-3.1-405B	Text Generation	Updated 88 days ago	1.15k	3.78k	



TRANSFORMER MODELS AND TASKS

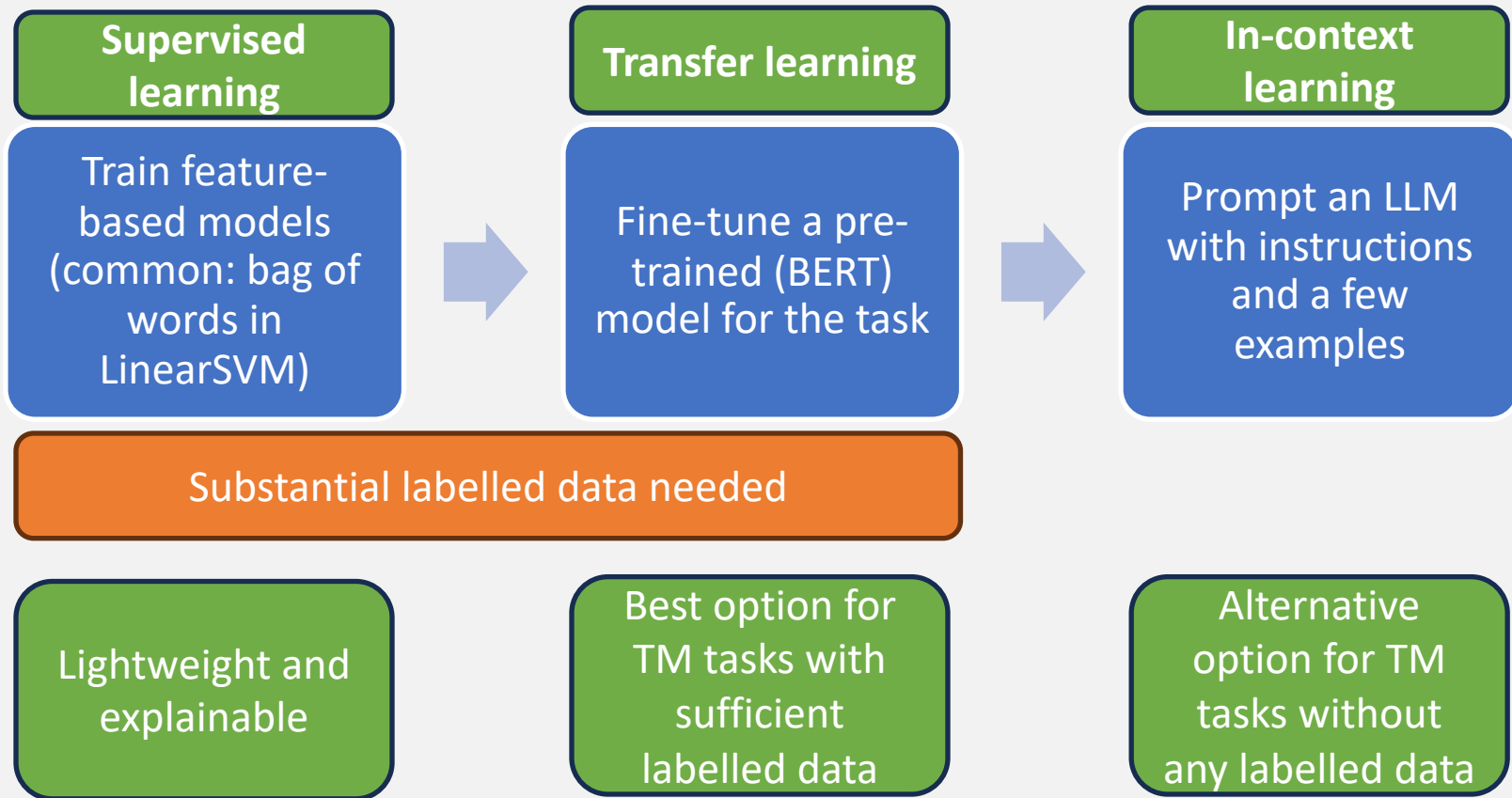
- **Encoder** models (BERT and family): classification (e.g., sentiment), sequence labelling (e.g. named entity recognition)
- **Decoder** models (GPT and family): text generation
- **Encoder-decoder** models (T5 and others): summarization, translation

The rise of the **Large Language Models (LLMs)**: when decoder models become HUGE (e.g. GPT-4), they can do sequence-to-sequence tasks (e.g. summarization) and even classification tasks (e.g. sentiment), by instructing them well in the prompt



More about LLMs
in week 6 and 7

3 MACHINE LEARNING PARADIGMS



Brown et al. (2020). Language models are few-shot learners
Wei et al. (2022). Emergent abilities of large language models

CASE

CASE: EXERCISE

- Goal: to discover side effects for hypertension medications
- Data:
 - 39,892 messages from a patient discussion forum on hypertension
- **How would you address this problem?**
Discuss in small groups

worsening symptoms since starting medication

Follow

Posted 2 weeks ago, 5 users are following.



dean89033

Hi all. I started out at 184/100, and was put on 40mg Lisinopril in January. Almost immediately I started to suffer with [vertigo](#) and dizzy spells where my hearing would cut out, cold sweats and fatigue. I knew that the first two weeks there would be some expected side effects, so I waited to see if they would pass. About a month later, I went back to my doctor, with a BP reading of 122/78, and she swapped me to Losartan Potassium 25mg. The same side effects continued, and after another month or so I went back (BP 126/90) and was swapped to Amlodipine Besylate 5mg. I've been on that since March, as there aren't any other medications my doctor can switch me to without going to beta blockers. But my symptoms have worsened.

It's like spacing out but worse? But also not like dissociating. For 10 or 15 seconds, I'm not "there" but I have the after image of whatever I was looking at before. Sometimes my eyes cross and I can't un-cross them, or sometimes they close and I can't keep them open, what normally brings me "back" is that I'll sway too much to one side, or my head will jerk down. It used to be that parts of my body would jerk but not so much now. I wouldn't say I'm confused? I know who/what/where I am but I don't know what I was doing/am doing/should do next. It doesn't feel like falling asleep, it feels like my brain lagged out, or I'm behind a loading screen in a video game. I've been tracking them and can't find any triggers.

I've had an MRI and EEG, and I'm waiting for the follow-up appointment in September to go over those results. I've also got a consultation with a cardiologist in October. I got a CPAP machine a few months ago with a diagnose of sleep apnea, but with the meds and stress from my symptoms it's hard to say if that's helped my BP any. We're trying to be thorough and check all the bases, but I

CASE: THE TEXT MINING PIPELINE

- Filter the data (retrieve relevant messages)
- Process the data (clean, anonymize)
- Create training data (human labelling) OR create an instruction prompt
- Identify medication names (named entity recognition)
- Identify side effects (named entity recognition)
- External knowledge needed (linking to a knowledge base)
- Relations between medications and side effects (relation extraction)

EXTRACTING SIDE EFFECTS FROM PATIENT EXPERIENCES: RESULTS

Automatic Extraction of Patient-reported Adverse Drug Events from a GIST patient forum

Medication:

Imatinib

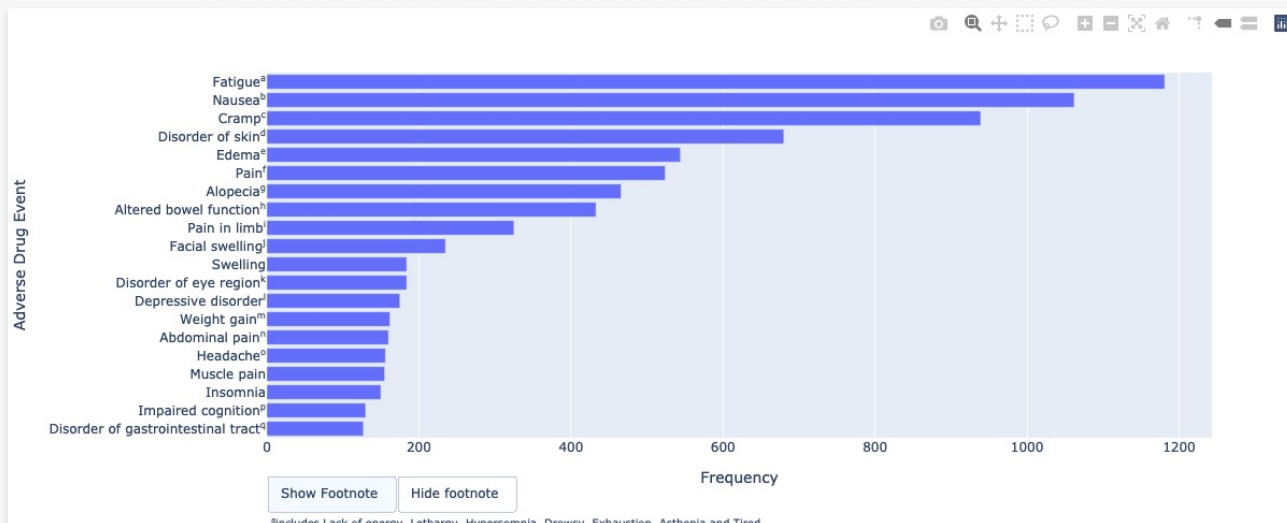
Type of analysis:

- ☒ Most prevalent ADE
☐ Long-term ADE
☐ Novel ADE

Amount of ADE displayed in:

- ☒ Frequency
☐ Percentage

WARNING! As the extraction was automatic, it may contain errors. These results were not manually verified.



Anne Dirkson, Suzan Verberne, Wessel Kraaij, Gerard van Oortmerssen and Hans Gelderblom. Automated gathering of real-world data from online patient forums can complement pharmacovigilance for rare cancers. Nature Scientific Reports 12, 10317 (2022). <https://doi.org/10.1038/s41598-022-13894-8>

COURSE STRUCTURE

COURSE GOALS

- <https://studiegids.universiteitleiden.nl/en/courses/123846/text-mining>
- You will learn about:
 - fundamentals of models (conceptual understanding)
 - practical applications
 - data, experimentation, evaluation

Admission requirements

Assumed prior knowledge

A Bachelor in AI or Computer Science is recommended for this course, as well as experience with programming in Python.

Description

Text mining is one of the core application areas of Natural Language Processing (NLP). Key text mining tasks are text classification, information extraction, and sentiment analysis. NLP is a fast developing field that is grounded in fundamental models for text representation. It has attracted much attention from researchers in other fields and the general public, especially in recent times with the increasing power of large language models (LLMs). This course gives an overview of the field from both a theoretical angle (underlying models) and a practical angle (applications, challenges with data). In addition to the lectures, the students work on practical assignments.

Outline:

Fundamentals:

week 1. Introduction
week 2. Text processing
week 3. Vector Semantics

Models and methods:

week 4. Text categorization
week 5. Data collection and annotation
week 6. Transformer models and transfer learning
week 7. Generative large language models

Applications:

week 8. Information Extraction
week 9. Topic Modelling & Text summarization
week 10. Sentiment analysis & stance detection
week 11. Industrial Text Mining (guest lecture)

week 12. Conclusions

Course objectives

After successful completion of this course, students are able to:

- explain the theoretical underpinnings and implementation of text representation models, both word based and embeddings based
- explain, both at the conceptual and the technical level, natural language processing (NLP) methods for sequence processing, in particular transformer models
- build models for a text mining task using machine learning algorithms and text data
- evaluate and report on given data, the developed models and methods
- reason, from a theoretical perspective, which models are applicable in which situations
- discuss which real-world challenges prevent the application of certain techniques, such as domain specificity, language variation, noise in the data, and trustworthiness of models
- explain and apply evaluation metrics and data quality metrics



COURSE LITERATURE

- The majority of the chapters come from this free, online book:
 - Dan Jurafsky and James H. Martin, Speech and Language Processing (3rd ed, August 2024) <https://web.stanford.edu/~jurafsky/slp3/>
- And a few papers / chapters from other sources
- The literature will be distributed on Brightspace, as are the slides

COURSE OUTLINE

➤ Course website: <http://tmr.liacs.nl/TM.html>

➤ No lecture on October 23

Week	Lecture	Literature	Exercise / assignment
1 (4 Sept)	Introduction		
2 (11 Sept)	Text processing	J&M chapter 2. Regular Expressions, Text Normalization, Edit Distance	Exercise: Chapter 1 of "Advanced NLP with Spacy"
3 (28 Sept)	Vector Semantics	(optional) J&M sections 7.1, 7.2, 7.3. Neural Networks J&M chapter 6. Vector Semantics	Exercise: Chapter 2 of "Advanced NLP with Spacy", sections 8-15
4 (25 Sept)	Text categorization	J&M chapter 4. Naive Bayes, Text Classification, and Sentiment	Exercise: Text classification tutorial (sklearn)
5 (2 Oct)	Data collection and annotation	TBD	Assignment 1. Text classification (deadline 8 October)
6 (9 Oct)	Transformer models and transfer learning	J&M chapter 9. Transformers J&M chapter 11 Masked Language Models	Exercise: Chapters 2 and 3 of the Huggingface NLP course
7 (16 Oct)	Generative large language models	J&M chapter 10. Large Language Models	
(23 Oct)	No lecture		
8 (30 Oct)	Information Extraction	J&M chapter 17. Sequence Labeling for Parts of Speech and Named Entities	Exercise: Token classification tutorial in the Huggingface NLP course
9 (6 Nov)	Topic Modelling & Text summarization	J&M chapter 12. Model Alignment, Prompting, and In-Context Learning	Exercise: Summarization tutorial in the Huggingface NLP course
10 (13 Nov)	Sentiment analysis & Stance detection	TBA	Assignment 2. Information Extraction (deadline 11 November)
11 (20 Nov)	Industrial Text Mining		Paper reading for the final assignment
12 (27 Nov)	Exam preparation session		
13 (4 Dec)	Online lab session		Final assignment (deadline 5 January)
(18 Dec)	Written exam		

GENERAL STRUCTURE

- 12 lecture weeks
- Homework:
 - **Literature** after the lecture
 - In some weeks you work on a **practical exercise** (online tutorial)
 - In other weeks you work on an **assignment** that you need to submit (2 smaller assignments, and one large assignment), in groups of 2
 - The assignments build on the tutorials so stay on schedule
- **Group work** is an integral part of the course. You will be expected to complete the assignments together with a team mate.

EXAM AND GRADE

- The assessment of the course consists of
 - a written exam (50% of course grade)
 - practical assignments (50% of course grade), in groups of 2
 - Assignment 1 (10%): text classification
 - Assignment 2 (10%): information extraction
 - Assignment 3 (30%): multiple topics to choose from
- Passing the course:
 - The grade for the written exam should be 5.5 or higher to pass the course.
 - The weighted average grade for the practical assignments should be 5.5 or higher to pass the course.
 - If a task is not submitted the grade for that task is 0.

DEADLINES

- All assignments will be submitted and graded through Brightspace.
- A TA will provide you with feedback
- Each assignment has a re-take opportunity,
 - but when submitted after the first deadline your maximum grade is 6

	Deadline	Re-sit deadline
Assignment 1	8 October	5 January (maximum grade 6)
Assignment 2	11 November	5 January (maximum grade 6)
Written exam	18 December	29 January
Final assignment	5 January	5 February (maximum grade 6)

CONCLUSIONS

HOMEWORK

- **Form groups for the practical assignments**
 - Make groups of 2 by first finding a teammate
 - (There is a discussion forum on Brightspace to help you if needed)
 - Enroll in a group on Brightspace (Groups -> Groups for assignments) with your teammate
- You can only see the assignments when (1) I published them and (2) you enrolled in a group

AFTER THIS LECTURE...

- You know what to expect from this course (both content and structure)
- You can explain the relation between text mining and **natural language processing**
- You can explain the relation between text mining and **information retrieval**
- You can describe the **text mining pipeline** on a general level
- You can list and explain the most important **challenges** of text data
- You can identify and explain tasks that represent text as **classification** object and tasks that represent text as **sequence**
- You can name a **transformer encoder** model and a task that you can solve with an encoder model
- You can name a **transformer decoder** model and a task that you can solve with a decoder model